

II-Probabilistic Approaches Probability, Marginal and Conditional Distributions

Foundations of Neuro-Symbolic AI

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Outline

Toy accounting example

Let us imagine that each possible assignment to is of equal frequency.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} [X_{A1}, X_{A2}, X_F]$$

We can ask the following questions: Is the account **Account 1** or the account **Account 2** more likely to be booked?

Since logic can only handle certainty, it cannot answer this question.

However, we can intuitively assign probabilities to the states:

$$\begin{pmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \frac{1}{3} & 0 \end{pmatrix} [X_{A1}, X_{A2}, X_F]$$

There are two worlds where **Account 1** is booked, and one world where **Account 2** is booked, thus

$$\mathbb{P}[X_{A1} = 1] = \frac{2}{3} > \mathbb{P}[X_{A2} = 1] = \frac{1}{3}.$$

Uncertainty: Summarizing the unknown

Uncertainties need to be handled:

- ▶ Not all variables are measured.

Example: When we do not know the value of **Feature**, then we do not know whether **Account 1** or **Account 2** have to be booked.

- ▶ Parts of the models are not known.

Example: Even if **Feature** is known to be **False**, then we do not know whether **Account 1** or **Account 2** have to be booked.

Outline

Probability distributions

Instead of storing 0/1 for possibilities, we can store probabilities (in the interval $[0, 1]$) in the coordinates of a tensor.

The total probability of all states must be 1 and can be calculated by contracting all legs of the tensor since

$$\sum_{\mathbf{x}_{[d]}} \mathbb{P} [X_{[d]} = \mathbf{x}_{[d]}] = \langle \mathbb{P} [X_{[d]}] \rangle_{[\emptyset]} .$$

Definition (Probability Tensor)

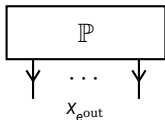
Let there be a factored system $X_{[d]}$ defined by variables X_k taking values in $[m_k]$. A probability distribution over the states of $X_{[d]}$ is a tensor $\mathbb{P} [X_{[d]}]$ with coordinates in the interval $[0, 1]$ such that

$$\langle \mathbb{P} [X_{[d]}] \rangle_{[\emptyset]} = 1 .$$

Directed hyperedges

Let us introduce **directed hyperedges**: Instead of just being subsets of the variables, directed hyperedges are pairs of incoming variables $e^{\text{in}} \subset \mathcal{V}$ and outgoing variables $e^{\text{out}} \subset \mathcal{V}$.

To highlight the normalization constraint we understand probability distributions as tensors on directed hyperedges, where all variables are outgoing:



Incoming variables will be used later to highlight conditioning.

Outline

Example: Being at a dentist

We are reasoning about a factored system with three variables:

- ▶ **Toothache** $x_0 \in [2]$, whether your tooth hurts
- ▶ **Cavity** $x_1 \in [2]$, whether there is a cavity in your tooth
- ▶ **Catch** $x_2 \in [2]$, whether the dentist catches in your tooth

Example: Being at a dentist

Let us take the following probability distribution over the states of the system:

Toothache	x_0	Cavity	x_1	Catch	x_2	$\mathbb{P}[X_0 = x_0, X_1 = x_1, X_2 = x_2]$
0		0		0		0.576
0		0		1		0.144
0		1		0		0.008
0		1		1		0.072
1		0		0		0.064
1		0		1		0.016
1		1		0		0.012
1		1		1		0.108

We align the probability values in the coordinates of an order-3 tensor:

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

Marginal distributions

Example: What is the relation between **Toothache** and **Cavity**?

- ▶ The variable **Catch** is not relevant for this question, thus we want to **marginalize** over it.
- ▶ To this end, we sum the probability of states differing in the value of **Catch** only.

This amounts to a contraction of the probability distribution tensor closing the variable X_2 :

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

↓

$$\begin{pmatrix} * & * \\ * & * \end{pmatrix} [X_0, X_1]$$

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$$\begin{pmatrix} 0.72 & * \\ * & * \end{pmatrix} [X_0, X_1]$$

Marginal distributions

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Marginal distributions

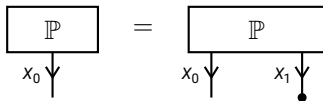
Summation of probabilities of states are done by contractions of the probability distribution tensor.

Definition (Marginal distribution)

Given a probability distribution $\mathbb{P} [X_{[d]}]$ and a subset $\mathcal{U} \subset [d]$, the marginal distribution of the variables $X_{\mathcal{U}}$ is the distribution

$$\mathbb{P} [X_{\mathcal{U}}] = \langle \mathbb{P} [X_{[d]}] \rangle_{[X_{\mathcal{U}}]} .$$

Example: The marginal distribution of $\mathcal{U} = \{X_0\}$ given a joint distribution of two variables X_0 and X_1 is the contraction



Outline

Example: Conditional distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states:

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} * & * \\ * & * \end{pmatrix} [X_0|X_1]$$

This is called a **conditional** distribution.

Example: Conditional distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states:

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} \frac{0.72}{0.72+0.08} & \frac{0.08}{0.72+0.08} \\ * & * \end{pmatrix} [X_0|X_1]$$

This is called a **conditional** distribution.

Example: Conditional distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states:

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \longrightarrow \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix} [X_1|X_0]$$

This is called a **conditional** distribution.

Normalizations

More general, conditionalization amounts to normalization of probability tensors.

Definition (Normalization)

The **normalization** of a tensor $\tau [X_{[d]}]$ with respect to disjoint subsets $\mathcal{V}^{\text{out}}, \mathcal{V}^{\text{in}} \subset [d]$ is the tensor

$$\langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}]}$$

defined by the coordinates

$$\begin{aligned} & \langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} = x_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}] } \\ &= \begin{cases} \frac{\langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} = x_{\mathcal{V}^{\text{out}}}, X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}]}}{\langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}]}} & \text{if } \langle \tau [X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{in}}} = x_{\mathcal{V}^{\text{in}}}]} \neq 0 \\ \frac{1}{\prod_{v \in \mathcal{V}^{\text{out}}} m_v} & \text{else} \end{cases} \end{aligned}$$

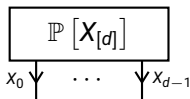
Normalized tensors and directed notation

Directed tensors represent normalizations: Each slice of the incoming variables is a normalized tensor, i.e. coordinates summing to 1.

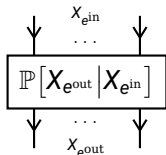
For example a probability distribution is normalized without incoming variables since

$$\langle \mathbb{P} [X_{[d]}] \rangle_{[X_{[d]}|\emptyset]} = \mathbb{P} [X_{[d]}]$$

This is denoted by



A conditional distribution is invariant under further normalizations (with the same incoming variables) and is thus denoted by

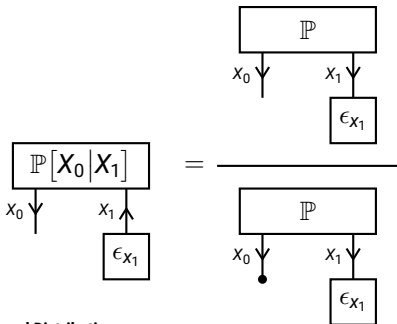


Formal Definition of Conditional Distributions

Definition (Conditional Probability)

Given a distribution $\mathbb{P}[X_{[d]}]$ and disjoint subsets $\mathcal{V}^{\text{out}}, \mathcal{V}^{\text{in}} \subset [d]$, the distribution of $X_{\mathcal{V}^{\text{out}}}$ conditioned on $X_{\mathcal{V}^{\text{in}}}$ is the normalization

$$\mathbb{P}[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}] = \langle \mathbb{P}[X_{[d]}] \rangle_{[X_{\mathcal{V}^{\text{out}}} | X_{\mathcal{V}^{\text{in}}}]}$$



Outline

Events and their probability

Events are subsets

$$\mathcal{U} \subset \times_{k \in [d]} [m_k]$$

of the possible samples. When the leg dimensions are 2 we build the formula

$$f[X_{[d]}] = \sum_{x_{[d]} \in \mathcal{U}} \epsilon_{x_{[d]}} [X_{[d]}] .$$

Probability of the event is then the contraction

$$\sum_{x_{[d]} : f[X_{[d]}=x_{[d]}]=1} \mathbb{P}[X_{[d]} = x_{[d]}] = \langle \mathbb{P}[X_{[d]}] , f[X_{[d]}] \rangle_{[\emptyset]} .$$

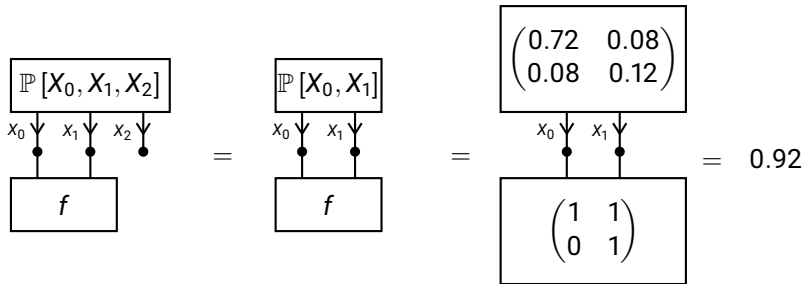
Toothache example

Example: What is the probability that there is a cavity whenever having toothache?

This is the event corresponding with the formula

$$f[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1]$$

Since only two variables are affected by this, the probability of the event can be answered by the marginal distribution:



Comparing logics and probability theory

Modelling assumptions

- ▶ **Ontological:** What properties does a system have?
→ We assume **fixed sets of categorical properties**.
- ▶ **Epistemological:** What can we tell about these properties?
→ We associate **possibilities or probabilities** with states.

Two main frameworks differing in the **empistemological** assumptions:

- ▶ Logics (**Possibilities**): The traditional mathematical model of intelligence.
- ▶ Probability Theory (**Probabilities**): Handling uncertainties about the systems state.

In both cases, we encode our knowledge in an arrangement of numbers:
Tensors with efficient representations by **Tensor Networks**.